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Power supply control device, in-vehicle control device, and power supply control method

Abstract

An individual ECU controls power supply through a wire W. A microcomputer of the individual ECU increases an average value of a wire current value of a current flowing through the wire W in a stepwise manner. The microcomputer determines, every time that the average value of a wire current value is increased, whether or not an anomaly has occurred in power supply through the wire (W) based on a temperature difference between a wire temperature of the wire (W) and an environmental temperature in an area surrounding the wire (W).

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5898557	12/1998	Baba	361/103	H03K 17/18
10166939	12/2018	Nakamura	N/A	G01K 7/42
10259407	12/2018	Iwasaki	N/A	B60R 16/03
11940334	12/2023	Sawano	N/A	G01K 7/22
2018/0294635	12/2017	Kozuki	N/A	H02H 3/085
2019/0165564	12/2018	Fuseya	N/A	H02H 5/047
2019/0312501	12/2018	Yamada	N/A	H02M 1/08
2021/0325259	12/2020	Sawano	N/A	B60R 16/02
2021/0391788	12/2020	Sawano et al.	N/A	N/A
2022/0045502	12/2021	Sawano et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2000-016200	12/1999	JP	N/A

OTHER PUBLICATIONS

International Search Report, Application No. PCT/JP2021/038412, mailed Dec. 28, 2021. ISA/Japan Patent Office. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is the U.S. national stage of PCT/JP2021/038412 filed on Oct. 18, 2021, which claims priority of Japanese Patent Application No. JP 2020-180894 filed on Oct. 28, 2020, the contents of which are incorporated herein.
TECHNICAL FIELD
(2) The present disclosure relates to a power supply control device, an in-vehicle control device, and a power supply control method.
BACKGROUND
(3) JP 2000-16200A discloses a power supply control device for a vehicle that controls power supply to a load. In this power supply control device, an FET (Field Effect Transistor) serving as a switch is disposed on a current path of a current flowing through the load. The power supply to the load is controlled by switching the FET on or off. When the temperature of the FET has increased to a temperature that is a predetermined temperature or more, while the FET is on, the FET is forcibly switched off. Accordingly, the temperature of the FET is prevented from increasing to an abnormal temperature.
(4) In a known power supply control device as described in JP 2000-16200A, when a cut-off condition for forcibly cutting off the current flow to a load is satisfied, the FET is forcibly switched off. When power supply to the load is started after the cut-off condition has been satisfied, the FET is again switched on, and the FET is kept on. When the FET is kept on in a state in which the factor that caused the cut-off condition to be satisfied is not resolved, the cut-off condition is satisfied again, and as a result, the FET is forcibly switched off again.
(5) The state in which the cut-off condition is satisfied is a state in which the current value of a current flowing through the FET is large, and is not preferable for the components that constitute the power supply control device. Therefore, an anomaly in the power supply needs to be detected while the current value is small.
SUMMARY
(6) The present disclosure aims to provide a power supply control device, an in-vehicle control device, and a power supply control method with which an anomaly in the power supply can be detected while the current value is small.
(7) A power supply control device according to one aspect of the present disclosure is a power supply control device that controls power supply through a wire, including a processing unit configured to execute processing, wherein the processing unit increases an average value of a wire current value of a current flowing through the wire in a stepwise manner, and every time that the average value of a wire current value is increased, determines whether or not an anomaly occurs in power supply through the wire based on a temperature difference between a wire temperature of the wire and an environmental temperature in an area surrounding the wire.
(8) An in-vehicle control device according to one aspect of the present disclosure is an in-vehicle control device that controls operations of a load, including: a receiving unit configured to receive instruction data instructing that the load be caused to start or stop operating; and a processing unit configured to execute processing, wherein the processing unit controls power supply to the load through a wire in accordance with instruction data received by the receiving unit, increases an average value of a wire current value of a current flowing through the wire in a stepwise manner, and determines, every time the average value of a wire current value is increased, whether or not an anomaly occurs in power supply through the wire, based on a temperature difference between a wire temperature of the wire and an environmental temperature in an area surrounding the wire.
(9) A power supply control method according to one aspect of the present disclosure is a power

supply control method of controlling power supply through a wire that includes steps to be executed by a computer, the method including: a step of increasing, in a stepwise manner, an average value of a wire current value of a current flowing through the wire; and a step of determining, every time the average value of a wire current value is increased, whether or not an anomaly occurs in power supply through the wire, based on a temperature difference between a wire temperature of the wire and an environmental temperature in an area surrounding the wire.
(10) Note that the present disclosure can be realized not only as a power supply control device that includes such characteristic processing units, but also as a power supply control method that includes the characteristic processing as steps, or as a computer program for causing a computer to execute these steps. Also, the present disclosure can be realized as a semiconductor integrated circuit that realizes all or part of the power supply control device, or as a power supply control system that includes the power supply control device.
Advantageous Effects of Disclosure
(11) According to the present disclosure, an anomaly in a power supply can be detected before the cut-off condition for cutting off the power supply is satisfied.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a block diagram illustrating the main configuration of a control system **1** in a first embodiment.
(2) FIG. **2** is a block diagram illustrating the main configuration of an individual ECU.
(3) FIG. **3** is a circuit diagram illustrating a switching device.
(4) FIG. **4** is a block diagram illustrating the main configuration of a microcomputer.
(5) FIG. **5** is a diagram illustrating content of a temperature difference table.
(6) FIG. **6** is a diagram illustrating content of an increasing amount table.
(7) FIG. **7** is a flowchart illustrating a procedure of temperature calculation processing.
(8) FIG. **8** is a flowchart illustrating a procedure of power supply control processing.
(9) FIG. **9** is a flowchart illustrating a procedure of power supply control processing.
(10) FIG. **10** is a state transition diagram of power supply.
(11) FIG. **11** is a diagram for describing a step-wise increase in the duty ratio of a PWM signal.
(12) FIG. **12** is a flowchart illustrating a procedure of power supply control processing in a second embodiment.
DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS
(13) First, modes for carrying out the disclosure will be enumerated and described. At least some of the modes described below may be combined as necessary.
(14) First Aspect
(15) In accordance with a first aspect, a power supply control device according to one aspect of the present disclosure is a power supply control device that controls power supply through a wire, and includes a processing unit configured to execute processing, wherein the processing unit increases an average value of a wire current value of a current flowing through the wire in a stepwise manner, and every time that the average value of a wire current value is increased, determines whether or not an anomaly has occurred in power supply through the wire based on a temperature difference between a wire temperature of the wire and an environmental temperature in an area surrounding the wire.
(16) Second Aspect
(17) In a second aspect, in the power supply control device according to one aspect of the present disclosure, the processing unit acquires the wire current value, calculates the temperature difference based on the acquired wire current value, and every time that the average value of the wire current

value is increased, determines whether or not the anomaly has occurred based on a calculated temperature difference.
(18) Third Aspect
(19) In a third aspect, the power supply control device according to one aspect of the present disclosure further includes: a switch disposed on a current path of current flowing through the wire; and a switching circuit configured to switch the switch on or off, wherein the processing unit causes the switching circuit to perform PWM control in which the switch is alternately switched on and off, and increases the average value of a wire current value in a stepwise manner by increasing a duty ratio in the PWM control in a stepwise manner.
(20) Fourth Aspect
(21) In a fourth aspect, the power supply control device according to one aspect of the present disclosure further includes: a storage unit that stores a plurality of upper limit values for the temperature difference in association with a plurality of duty ratios regarding the PWM control, wherein the processing unit determines, every time the average value of a wire current value is increased, whether or not the anomaly occurs, based on whether or not the temperature difference exceeds an upper limit value associated with a duty ratio in the PWM control performed by the switching circuit.
(22) Fifth Aspect
(23) In a fifth aspect, in the power supply control device according to one aspect of the present disclosure, the processing unit determines, every time the average value of a wire current value is increased, whether or not the anomaly has occurred, based on the amount of increase in the temperature difference that has increased due to the increase in the average value of a wire current value.
(24) Sixth Aspect
(25) In a sixth aspect, the power supply control device according to one aspect of the present disclosure further includes: a switch to be disposed on a current path of current flowing through the wire; and a switching circuit configured to switch the switch on or off, wherein the switching circuit receives a signal, the switching circuit switches the switch on or off according to the received signal, the switching circuit, if the wire current value is a current threshold or more, or if the temperature of the switch is a switch temperature threshold or more, switches the switch off irrelevant to the received signal, and the processing unit increases the average value of a wire current value in a stepwise manner, after the switching circuit has switched the switch off irrelevant to the received signal.
(26) Seventh Aspect
(27) In a seventh aspect, in the power supply control device according to one aspect of the present disclosure, the processing unit cuts off a flow of current flowing through the wire, when the wire temperature is a wire temperature threshold or more, and increases the average value of a wire current value in a stepwise manner, after the wire temperature has increased to a temperature that is the wire temperature threshold or more.
(28) Eighth Aspect
(29) In an eighth aspect, an in-vehicle control device according to one aspect of the present disclosure is an in-vehicle control device that controls operations of a load, and includes: a receiving unit configured to receive instruction data instructing that the load be caused to start or stop operating; and a processing unit configured to execute processing, wherein the processing unit controls power supply to the load through a wire in accordance with instruction data received by the receiving unit, increases an average value of a wire current value of a current flowing through the wire in a stepwise manner, and determines, every time the average value of a wire current value is increased, whether or not an anomaly has occurred in power supply through the wire, based on a temperature difference between a wire temperature of the wire and an environmental temperature in an area surrounding the wire.

(30) Ninth Aspect
(31) In a ninth aspect, a power supply control method according to one aspect of the present disclosure is a power supply control method of controlling power supply through a wire that includes steps to be executed by a computer, and the method includes: a step of increasing, in a stepwise manner, an average value of a wire current value of a current flowing through the wire; and a step of determining, every time the average value of a wire current value is increased, whether or not an anomaly has occurred in power supply through the wire, based on a temperature difference between a wire temperature of the wire and an environmental temperature in an area surrounding the wire.
(32) In the power supply control device, in-vehicle control device, and power supply control method according to one aspect described above, an average value of a wire current value (average wire current value) in a given period is increased in a stepwise manner. Every time the average wire current value is increased, whether or not an anomaly has occurred in power supply is determined based on a temperature difference between the wire temperature and the environmental temperature. Therefore, an anomaly in power supply can be detected while a current value is small.
(33) In the power supply control device according to one aspect described above, the temperature difference between the wire temperature and the environmental temperature is calculated based on a wire current value. The calculated temperature difference is used to determine whether or not an anomaly has occurred in power supply.
(34) In the power supply control device according to one aspect described above, the average wire current value is increased in a stepwise manner by increasing the duty ratio in PWM control performed on a switch in a stepwise manner. Therefore, the average wire current value can be easily increased in a stepwise manner.
(35) In the power supply control device according to one aspect described above, determination as to whether or not an anomaly has occurred in power supply is made based on a comparison between the temperature difference and the upper limit value associated with the duty ratio in PWM control actually performed by the switching circuit.
(36) In the power supply control device according to one aspect described above, determination as to whether or not an anomaly has occurred in power supply is made based on the amount of increase in the temperature difference caused by the increase in duty ratio in PWM control.
(37) In the power supply control device according to one aspect described above, the switching circuit forcibly switches off the switch based on a wire current value or a switch temperature. After the switching circuit forcibly switches off the switch, the average wire current value is increased in a stepwise manner, and whether or not an anomaly has occurred in power supply is determined.
(38) In the power supply control device according to one aspect described above, when the wire temperature has increased to a temperature that is the wire temperature threshold or more, the current flow through the wire is forcibly cut off. After the current flow is forcibly cut off, the average wire current is increased in a stepwise manner, and whether or not an anomaly has occurred in power supply is determined.
(39) Specific examples of a control system according to embodiments of the disclosure will be described below with reference to the drawings. Note that the present disclosure is not limited to these illustrative examples and is defined by the claims, and all changes that come within the meaning and range of equivalency of the claims are intended to be embraced therein.
First Embodiment
(40) Configuration of Control System
(41) FIG. **1** is a block diagram illustrating the main configuration of a control system **1** in a first embodiment. The control system **1** is mounted in a vehicle C. The control system **1** includes an integrated ECU **10**, an individual ECU **11***a*, a plurality of individual ECUs **11***b*, a DC power supply **12**, a load **13**, an actuator **14**, and two sensors **15***a* and **15***b*. The DC power supply **12** is a battery, for example. In FIG. **1**, interconnect lines for supplying power are shown with thick lines.

Interconnect lines through which data or signals propagates are shown with thin lines.
(42) The integrated ECU **10** is connected to the individual ECU **11***a* and the plurality of individual ECUs **11***b*. The individual ECU **11***a* is connected to a positive electrode of the DC power supply **12** and one end of a wire W. The other end of the wire W is connected to one end of the load **13**. A negative electrode of the DC power supply **12** and the other end of the wire W are grounded. The sensor **15***a* is also connected to the individual ECU **11***a*. The actuator **14** and the sensor **15***b* are connected to each individual ECU **11***b*.
(43) The DC power supply **12** supplies power to the individual ECU **11***a*, and also supplies power to the load **13** through the wire W. The individual ECU **11***a* controls power supply to the load **13** through the wire W. The individual ECU **11***a* functions as a power supply control device. The load **13** is an electrical device such as a lamp or a motor. When power is supplied to the load **13**, the load **13** starts operating. When power supply to the load **13** stops, the load **13** stops operating. The individual ECU **11***a* controls operations of the load **13** by controlling the power supply to the load **13**. The individual ECU **11***a* also functions as an in-vehicle control device.
(44) The actuator **14** is also an electrical device. The individual ECU **11***b* outputs a control signal indicating operations to be performed by the actuator **14** to the actuator **14**. Upon receiving a control signal, the actuator **14** performs operations indicated by the received control signal.
(45) The sensors **15***a* and **15***b* repeatedly generate vehicle data regarding the vehicle C. The vehicle data is image data obtained by capturing a surrounding area of the vehicle C, data indicating the speed of the vehicle C, data indicating whether or not a switch mounted on the vehicle C is on, or the like. Every time the sensor **15***a* generates vehicle data, the sensor **15***a* outputs the generated vehicle data to the individual ECU **11***a*. Similarly, every time the sensor **15***b* generates vehicle data, the sensor **15***b* outputs the generated vehicle data to the individual ECU **11***b*. Every time the individual ECUs **11***a* and **11***b* receive vehicle data, the individual ECUs **11***a* and **11***b* transmit the received vehicle data to the integrated ECU **10**.
(46) The integrated ECU **10** determines the operations of the load **13** based on one or more pieces of vehicle data received from at least one of the individual ECU **11***a* and the plurality of individual ECUs **11***b*. Here, the operations of the load **13** to be determined are to start operating or to stop operating. Upon determining the operations of the load **13**, the integrated ECU **10** transmits instruction data for instructing the determined operations to the individual ECU **11***a*. Upon receiving instruction data from the integrated ECU **10**, the individual ECU **11***a* causes the load **13** to perform the operations as instructed by the received instruction data.
(47) Similarly, the integrated ECU **10** determines the operations of one or more actuators **14** based on one or more pieces of vehicle data received from at least one of the individual ECU **11***a* and the plurality of individual ECUs **11***b*. Upon determining the operations of the one or more actuators **14**, the integrated ECU **10** transmits instruction data for instructing the determined operations to the corresponding one or more individual ECUs **11***b*. Upon receiving instruction data from the integrated ECU **10**, the individual ECU **11***b* outputs a control signal to the actuator **14** connected to the individual ECU **11***b*. The operations indicated by the control signal are operations instructed by the instruction data received by the individual ECU **11***b*. As mentioned above, the actuator **14** performs operations indicated by the received control signal.
(48) Configuration of Individual ECU **11***a*
(49) FIG. **2** is a block diagram illustrating the main configuration of the individual ECU **11***a*. The individual ECU **11***a* includes a switching device **20**, a microcomputer **21**, a voltage detecting unit **22**, and an environmental temperature detecting unit **23**. The switching device **20** is connected between the positive electrode of the DC power supply **12** and one end of the wire W. The switching device **20** is also connected to the microcomputer **21**. The voltage detecting unit **22** is connected to the positive electrode of the DC power supply **12**. The voltage detecting unit **22** and the environmental temperature detecting unit **23** are separately connected to the microcomputer **21**. The microcomputer **21** is also connected to the integrated ECU **10** and the sensor **15***a*.

(50) The switching device **20** includes a switch **30** (see FIG. **3**). The switch **30** is disposed on a current path of a current flowing from the positive electrode of the DC power supply **12** to the load **13**. When the switch **30** is switched on, a current flows from the positive electrode of the DC power supply **12** to the switch **30**, the wire W, and the load **13** in this order. With this, power is supplied to the load **13**. When the switch **30** is switched off, the current flow is cut off, and the power supply to the load **13** is stopped.
(51) The switching device **20** outputs analog current value information indicating a wire current value of a current flowing through the wire W, to the microcomputer **21**. The current value information is represented by a voltage proportional to the wire current. The switching device **20** also outputs switch temperature information indicating the temperature of the switch **30** to the microcomputer **21**. Below, the temperature of the switch **30** is denoted as a switch temperature. The switch temperature information is represented by a voltage that changes according to the switch temperature.
(52) The microcomputer **21** outputs a PWM (Pulse Width Modulation) signal or an off signal for causing the switch **30** to be off to the switching device **20**. The PWM signal includes high-level voltage periods and low-level voltage periods. The off signal is at a low-level voltage. In the PWM signal, switching from a low-level voltage to a high-level voltage is periodically performed. The duty ratio of a PWM signal is a ratio of a period, in one cycle, in which the voltage of the PWM signal is at the high-level voltage. The unit of the duty ratio is percent. The duty ratio exceeds 0% and is 100% or less. The duty ratio is adjusted by adjusting the timing at which the signal is switched from the high-level voltage to the low-level voltage.
(53) Note that, in the PWM signal, the signal may also be periodically switched from the high-level voltage to the low-level voltage. In this case, the duty ratio is adjusted by adjusting the timing at which the signal is switched from the low-level voltage to the high-level voltage.
(54) Below, it is assumed that the microcomputer **21** is outputting a PWM signal to the switching device **20** in a state in which the wire current is less than a fixed current threshold, and the switch temperature is less than a fixed switch temperature threshold. When the PWM signal voltage is switched from a low-level voltage to a high-level voltage in this case, the switching device **20** switches the switch **30** from off to on. When the PWM signal voltage is switched from a high-level voltage to a low-level voltage in a similar case, the switching device **20** switches the switch **30** from on to off. When the microcomputer **21** is outputting a PWM signal, power is supplied to the load **13**.
(55) As described above, the switching device **20** performs PWM control for alternately switching switch **30** on and off according to the received PWM signal. The duty ratio in the PWM control indicates a ratio of a period, in a given period, in which the switch **30** is on. The duty ratio in the PWM control matches the duty ratio of the PWM signal. As the duty ratio of the PWM signal increases, the period in which the switch **30** is on increases. Therefore, as the duty ratio of the PWM signal increases, the average wire current value in a given period increases. The given period is one cycle of the PWM signal, for example.
(56) When the microcomputer **21** is outputting an off signal, the switching device **20** keeps the switch **30** off. Therefore, when the microcomputer **21** is outputting the off signal, the load **13** stops operating.
(57) When the wire current is the current threshold or more in the case where the microcomputer **21** is outputting a PWM signal to the switching device **20**, the switching device **20** forcibly switches off the switch **30** irrelevant to the PWM signal input to the switching device **20**, and keeps the switch **30** off. Below, the forcible switching off of the switch **30** is denoted as “self-shutting off”. As a result of forcibly switching the switch **30** off, the current flow through the wire W is forcibly cut off.
(58) When the wire current has increased to the current threshold or more, the switching device **20** outputs the largest voltage that is allowed to be output to the microcomputer **21**. Accordingly, the

execution of the self-shutting off is notified to the microcomputer **21**. Upon receiving notification of the execution of the self-shutting off, the microcomputer **21** outputs an off signal to the switching device **20**. When the microcomputer **21** outputs an off signal to the switching device **20**, the self-shutting off is canceled.
(59) When the witch temperature is the switch temperature threshold or more in the case where the microcomputer **21** is outputting a PWM signal to the switching device **20**, the switching device **20** performs self-shutting off of the switch **30**, irrelevant to the PWM signal input to the switching device **20**. When the switch temperature has increased to a temperature that is the witch temperature threshold or more, the switching device **20** outputs the largest voltage that is allowed to be output to the microcomputer **21**. Accordingly, the microcomputer **21** is notified of the execution of the self-shutting off. Upon receiving a notification indicating the execution of the self-shutting off, the microcomputer **21** outputs an off signal to the switching device **20**. When the microcomputer **21** outputs an off signal to the switching device **20**, the self-shutting off is canceled.
(60) The voltage detecting unit **22** detects the voltage between two ends of the DC power supply **12**. Below, the voltage between two ends of the DC power supply **12** is denoted as a power supply voltage. The voltage detecting unit **22** outputs analog power supply voltage information indicating the detected power supply voltage to the microcomputer **21**. The power supply voltage information is represented by a voltage obtained by voltage-dividing the power supply voltage, for example.
(61) The environmental temperature detecting unit **23** detects the environmental temperature in an area surrounding the wire W. The environmental temperature is an ambient temperature of the wire W. The environmental temperature detecting unit **23** outputs analog environmental temperature information indicating the detected environmental temperature. The environmental temperature information is represented by a voltage that changes according to the environmental temperature, for example.
(62) Every time the sensor **15***a* generates vehicle data, the sensor **15***a* outputs the generated vehicle data to the microcomputer **21**.
(63) The microcomputer **21** transmits the vehicle data received from the sensor to the integrated ECU **10**. The microcomputer **21** receives instruction data for instructing that the load **13** be caused to start or stop operating from the integrated ECU **10**. Upon receiving instruction data for instructing that the load **13** be caused to start operating, the microcomputer **21** outputs a PWM signal to the switching device **20**. When the microcomputer **21** outputs a PWM signal to the switching device **20**, the switching device **20** performs PWM control on the switch **30**, and power is supplied to the load **13**. As a result, the load **13** starts operating.
(64) Upon receiving instruction data for instructing that the load **13** be caused to stop operating, the microcomputer **21** outputs an off signal to the switching device **20**. Accordingly, power supply to the load **13** stops, and the load **13** stops operating. When outputting a PWM signal to the switching device **20**, the microcomputer **21** adjusts the duty ratio of the PWM signal based on the power supply voltage indicated by the power supply voltage information received from the voltage detecting unit **22**. Also, the microcomputer **21** repeatedly calculates the wire temperature of the wire W based on the wire current value of the wire W that is indicated by the current value information input from the switching device **20**, the environmental temperature indicated by the environmental temperature information received from the environmental temperature detecting unit **23**, and the duty ratio of the PWM signal output to the switching device **20**.
(65) When the calculated wire temperature of the wire W is a fixed wire temperature threshold or more, the microcomputer **21** outputs an off signal to the switching device **20**. With this, the switching device **20** switches the switch **30** off. As a result, the power supply to the load **13** stops, and the load **13** stops operating.
(66) Configuration of Switching Device **20**
(67) FIG. **3** is a circuit diagram of the switching device **20**. The switching device **20** includes a driving circuit **31**, a current detecting circuit **32**, and a switch temperature detecting circuit **33**, in

addition to the switch **30**. The switch **30** is an N-channel FET. The current detecting circuit **32** includes a current output unit **40** and a current detection resistor **41**. The switch temperature detecting circuit **33** includes an NTC (Negative Temperature Coefficient) type thermistor **50** and a temperature detection resistor **51**.

(68) The drain of the switch **30** is connected to the positive electrode of the DC power supply **12**. The source of the switch **30** is connected to one end of the wire W. As described above, the other end of the wire W is connected to one end of the load **13**. The gate of the switch **30** is connected to the driving circuit **31**. The driving circuit **31** is also connected to the microcomputer **21**.

(69) The drain of the switch **30** is also connected to the current output unit **40** of the current detecting circuit **32**. The current output unit **40** is also connected to one end of the current detection resistor **41**. The other end of the current detection resistor **41** is grounded. The connection node between the current output unit **40** and the current detection resistor **41** is connected to the microcomputer **21** and the driving circuit **31**.

(70) In the switch temperature detecting circuit **33**, a fixed voltage is applied to one end of the thermistor **50**. The fixed voltage is generated by a regulator (not shown) stepping down the voltage between two ends of the DC power supply **12**, for example. The reference potential of the fixed voltage is the ground potential. The voltage value of the fixed voltage is denoted as Vc. The other end of the thermistor **50** is connected to one end of the temperature detection resistor **51**. The other end of the temperature detection resistor **51** is grounded. The connection node between the thermistor **50** and the temperature detection resistor **51** is connected to the microcomputer **21** and the driving circuit **31**.

(71) In the switch **30**, when the gate voltage relative to the source potential is a given voltage or more, the switch **30** is on. When the switch **30** is on, the resistance between the drain and source, in the switch **30**, is sufficiently small. Therefore, a current can flow through the drain and source of the switch **30**. When the switch **30** is on, a current flows from the positive electrode of the DC power supply **12** to the switch **30**, the wire W, and the load **13**, in this order. Accordingly, the switch **30** is disposed on a current path of current flowing through the wire W.

(72) In the switch **30**, when the gate voltage relative to the source potential is less than the given voltage, the switch **30** is off. When the switch **30** is off, the resistance between the drain and source, in the switch **30**, is sufficiently large. Therefore, a current cannot flow through the drain and source of the switch **30**. When the switch **30** is off, a current cannot flow through the switch **30** and the wire W.

(73) The microcomputer **21** outputs a PWM signal to the driving circuit **31**. Assume that the wire current value is less than the current threshold, and the switch temperature is less than the switch temperature threshold. When the PWM signal voltage is switched from a low-level voltage to a high-level voltage in this case, the driving circuit **31** increases the gate voltage relative to the ground potential, in the switch **30**. Accordingly, in the switch **30**, the gate voltage relative to the source potential increases to a voltage that is the given voltage value or more, and the switch **30** is switched on.

(74) In a similar case, when the PWM signal voltage is switched from a high-level voltage to a low-level voltage, the driving circuit **31** reduces the gate voltage relative to the ground potential, in the switch **30**. Accordingly, in the switch **30**, the gate voltage relative to the ground potential decreases to a voltage that is less than the given voltage value, and the switch **30** is switched off.

(75) As described above, the driving circuit **31** switches the switch **30** on or off, by adjusting the gate voltage relative to the source potential. The driving circuit **31** functions as a switching circuit. Assume that the wire current is less than the current threshold, and the switch temperature is less than the switch temperature threshold. When the microcomputer **21** is outputting a PWM signal to the driving circuit **31** in this case, the driving circuit **31** performs PWM control on the switch **30** according to the PWM signal voltage. As described above, the duty ratio in the PWM control matches the duty ratio of the PWM signal. When the driving circuit **31** performs PWM control on

the switch **30**, power is supplied to the load **13**.

(76) The microcomputer **21** outputs an off signal to the driving circuit **31**. When the microcomputer **21** outputs an off signal to the driving circuit **31**, the driving circuit **31** switches the switch **30** off. While the microcomputer **21** is outputting an off signal, the driving circuit **31** keeps the switch **30** off.

(77) In the current detecting circuit **32**, the current output unit **40** draws in current from the drain of the switch **30**, and outputs the drawn-in current to the current detection resistor **41**. The current value of current drawn-in by the current output unit **40** is proportional to the wire current and is represented by a formula (wire current value)/(predetermined number). The predetermined number is 1000, for example. The wire current value is a current value of a current flowing through the switch **30** and the wire W.

(78) The current detecting circuit **32** outputs the voltage between two ends of the current detection resistor **41** to the microcomputer **21** and the driving circuit **31**, as the current value information. The current value information is represented by a voltage obtained using the formula (wire current value).Math.(resistance value of the current detection resistor **41**)/(predetermined number). The symbol ".Math." indicates multiplication. The resistance value of the current detection resistor **41** and the predetermined number are fixed values, and therefore the wire current value can be calculated based on the current value information. The current value information increases as the wire current increases.

(79) While the microcomputer **21** is outputting a PWM signal to the driving circuit **31**, a current flows through the switch **30** and a wire W. When the wire current value indicated by the current value information has increased to a current value that is the wire current threshold or more, the driving circuit **31** performs self-shutting off of the switch **30** irrelevant to the input PWM signal.

(80) When the wire current value indicated by the current value information has increased to a current value that is the wire current threshold or more, the driving circuit **31** applies a voltage between two ends of the current detection resistor **41**. This voltage is a largest voltage that is allowed to be output to the microcomputer **21**. The largest voltage is input to the microcomputer **21**. With this, the microcomputer **21** is notified of execution of self-shutting off of the driving circuit **31**. As described above, upon receiving a notification of execution of self-shutting off, the microcomputer **21** outputs an off signal to the driving circuit **31**. When the microcomputer **21** outputs an off signal, the self-shutting off of the driving circuit **31** is canceled.

(81) In the switch temperature detecting circuit **33**, the fixed voltage whose voltage value is Vc is voltage-divided by the thermistor **50** and the temperature detection resistor **51**. The switch temperature detecting circuit **33** outputs the divided voltage obtained by voltage-dividing the fixed voltage to the microcomputer **21** and the driving circuit **31**, as switch temperature information. The resistance value of the temperature detection resistor **51** is denoted as rd. The resistance value of the thermistor **50** is denoted as rt. The switch temperature information, that is, the divided voltage value is represented by a formula Vc.Math.rd/(rd+rt). The voltage value Vc and the resistance value rd are fixed values, and therefore the switch temperature information indicates the resistance value rt of the thermistor **50**.

(82) The thermistor **50** is an NTC type thermistor, and therefore the resistance value rt decreases as the temperature of the thermistor **50** increases. The thermistor **50** is disposed in the vicinity of the switch **30**. When the switch temperature of the switch **30** increases, the temperature of the thermistor **50** increases. When the switch temperature decreases, the temperature of the thermistor **50** decreases. Accordingly, the resistance value rt of the thermistor **50** decreases as the switch temperature increases. Therefore, the resistance value rt of the thermistor **50** indicates the switch temperature. The switch temperature information, that is, the divided-voltage increases as the switch temperature increases.

(83) When the switch temperature indicated by the switch temperature information has increased to a temperature that is the switch temperature threshold or more, the driving circuit **31** applies a

voltage between two ends of the temperature detection resistor **51**. This voltage is the largest voltage that is allowed to be output to the microcomputer **21**. The largest voltage is input to the microcomputer **21**. With this, the microcomputer **21** is notified of execution of self-shutting off of the driving circuit **31**. As described above, upon receiving a notification of execution of self-shutting off, the microcomputer **21** outputs an off signal to the driving circuit **31**. When the microcomputer **21** outputs an off signal, the self-shutting off of the driving circuit **31** is canceled.
(84) As described above, when the microcomputer **21** outputs a PWM signal to the driving circuit **31**, the driving circuit **31** performs PWM control on the switch **30**. As a result, a current flows through the switch **30** and the wire W, and the switch temperature increases. When the wire current has increased to the wire current threshold or more, or when the switch temperature has increased to a temperature that is the switch temperature threshold or more, the driving circuit **31** performs self-shutting off, and notifies the microcomputer **21** of the fact that self-shutting off has been executed. Upon receiving a notification of execution of self-shutting off, the microcomputer **21** outputs an off signal to the driving circuit **31**. With this, the self-shutting off of the driving circuit **31** is canceled.
(85) Note that, in the switch temperature detecting circuit **33**, the type of the thermistor **50** is not limited to NTC, and may also be PTC (Positive Temperature Coefficient). In this case, the resistance value of the thermistor **50** increases, as the temperature of the thermistor **50**, that is, the switch temperature increases. Therefore, the switch temperature information decreases as the switch temperature increases. Moreover, the positions of the thermistor **50** and the temperature detection resistor **51** may also be reversed. In this case, the thermistor **50** is grounded, and a fixed voltage is applied to the temperature detection resistor **51**. In the case where a fixed voltage is applied to the temperature detection resistor **51**, when the type of the thermistor **50** is NTC, the switch temperature information decreases as the switch temperature increases. When the type of the thermistor is PTC in a similar case, the switch temperature information increases as the switch temperature increases. In the case where a fixed voltage is applied to the temperature detection resistor **51**, when the driving circuit **31** has performed self-shutting off, the driving circuit **31** applies a voltage to the thermistor **50**.
(86) Main Configuration of Microcomputer **21**
(87) FIG. **4** is a block diagram illustrating the main configuration of the microcomputer **21**. The microcomputer **21** includes A/D conversion units **60**, **61**, **62**, and **63**, an output unit **64**, an input unit **65**, a communication unit **66**, a storage unit **67**, and a control unit **68**. These units are connected to an internal bus **69**. The A/D conversion units **60**, **61**, **62**, and **63** are respectively also connected to the voltage detecting unit **22**, the switch temperature detecting circuit **33**, the current detecting circuit **32**, and the environmental temperature detecting unit **23**. The output unit **64** is also connected to the driving circuit **31**. The input unit **65** is also connected to the sensor **15***a*. The communication unit **66** is also connected to the integrated ECU **10**.
(88) The voltage detecting unit **22** outputs analog power supply voltage information to the A/D conversion unit **60**. The A/D conversion unit **60** converts the received analog power supply voltage information to digital power supply voltage information. The control unit **68** acquires the digital power supply voltage information from the A/D conversion unit **60**.
(89) The output unit **64** is an interface. The output unit **64** outputs a PWM signal and an off signal to the driving circuit **31** in accordance with an instruction from the control unit **68**. The duty ratio of the PWM signal output by the output unit **64** is adjusted by the control unit **68**.
(90) The switch temperature detecting circuit **33** outputs analog switch temperature information to the A/D conversion unit **61**. The A/D conversion unit **61** converts the received analog switch temperature information to digital switch temperature information. The control unit **68** acquires the digital switch temperature information from the A/D conversion unit **61**.
(91) The current detecting circuit **32** outputs analog current value information to the A/D conversion unit **62**. The A/D conversion unit **62** converts the received analog current value

information to digital current value information. The control unit **68** acquires the digital current value information from the A/D conversion unit **62**.
(92) The environmental temperature detecting unit **23** outputs analog environmental temperature information to the A/D conversion unit **63**. The A/D conversion unit **63** converts the received analog environmental temperature information to digital environmental temperature information. The control unit **68** receives the digital environmental temperature information from the A/D conversion unit **63**.
(93) The input unit **65** is an interface. Every time the sensor **15***a* generates vehicle data, the sensor **15***a* outputs the generated vehicle data to the input unit **65**. The control unit **68** acquires the vehicle data received from the sensor **15***a* from the input unit **65**.
(94) The communication unit **66** transmits vehicle data to the integrated ECU in accordance with an instruction from the control unit **68**. The communication unit **66** receives instruction data for instructing that the load **13** be caused to start or stop operating from the integrated ECU **10**. The communication unit **66** functions as a receiving unit.
(95) The storage unit **67** is a nonvolatile memory. A computer program P is stored in the storage unit **67**. The control unit **68** includes a processing element that executes processing, such as a CPU (Central Processing Unit). The control unit **68** functions as a processing unit. The processing element (computer) of the control unit **68** executes, in parallel, vehicle data transmission processing, temperature calculation processing, power supply control processing, and the like, by executing the computer program P. The vehicle data transmission processing is processing for transmitting vehicle data to the integrated ECU **10**. The temperature calculation processing is processing for calculating the wire temperature of the wire W. The power supply control processing is processing for controlling power supply to the load **13**.
(96) Note that the computer program P may also be stored in a non-transitory storage medium A such that the processing element of the control unit **68** can read the computer program P. In this case, the computer program P read out from the storage medium A by a reading device (not shown) is written into the storage unit **67**. The storage medium A is an optical disk, a flexible disk, a magnetic disk, a magneto-optical disk, a semiconductor memory, or the like. The optical disk is a CD (Compact Disc)-ROM (Read Only Memory), a DVD (Digital Versatile Disc)-ROM, a BD (Blu-ray (registered trademark) Disc), or the like. The magnetic disk is a hard disk, for example. Also, the computer program P may also be downloaded from an external device (not shown) that is connected to a communication network (not shown), and the downloaded computer program P may be written into the storage unit **67**.
(97) The number of processing elements included in the control unit **68** is not limited to one, and may also be two or more. When the number of processing elements included in the control unit **68** is two or more, the plurality of processing elements execute the vehicle data transmission processing, the temperature calculation processing, the power supply control processing, and the like, in a cooperative manner.
(98) The control unit **68** periodically executes the temperature calculation processing. In the temperature calculation processing, the control unit **68** calculates the temperature difference between the wire temperature and the environmental temperature, and adds the environmental temperature to the calculated temperature difference. With this, the wire temperature is calculated.
(99) In the calculation of the wire temperature, the control unit **68** calculates a temperature difference ΔTw by substituting a prior temperature difference ΔTp that was calculated previously, a wire current Iw of the wire W, an environmental temperature Ta, and a duty ratio D of the PWM signal, that is, a duty ratio D in the PWM control, in the following formulas [1], [2].

$$\Delta Tw=\Delta Tp\cdot\exp(-\Delta t/\tau r)+Rth\cdot Rw\cdot D\cdot Iw^{2}\cdot(1-\exp(-\Delta t/\tau r))/100 \quad [1]$$

$$Rw=Ro\cdot(1+K(Ta+\Delta Tp-To)) \quad [2]$$

(100) The variables and constants used in the formulas [1] and [2] will be described. In the

description of the variables and constants, the units of the variables and constants are also shown. ΔTw, ΔTp, Ta, Iw, Rw, Rth, and D are respectively a calculated temperature difference (° C.), a prior temperature difference (° C.), an environmental temperature (° C.), a wire current (A) of the wire W, a wire resistance (Ω) of the wire W, a wire thermal resistance (° C./W) of the wire W, and a duty ratio of the PWM signal, as described above. Δt is a cycle (s) with which the temperature difference ΔTw is calculated, that is, a cycle with which the temperature calculation processing is executed. τr is a wire heat release time constant (s) of the wire W.

(101) To is a predetermined temperature (° C.). Ro is a wire resistance (Q) at the temperature To. K. is a wire resistance temperature coefficient (/° C.) of the wire W. The temperature difference ΔTw, prior temperature difference ΔTp, wire current Iw, and environmental temperature Ta are variables. The cycle Δt, wire heat release time constant τr, wire thermal resistance Rth, wire resistance Ro, wire resistance temperature coefficient K, and temperature To are preset constants. The value of the first term in the formula [1] decreases as the cycle Δt increases, and therefore the first term of the computation formula [1] represents heat released from the wire W Also, the value of the second term of the formula [1] increases as the cycle Δt increases, and therefore, the second term of the formula [1] represents the heat generated by the wire W.

(102) The wire temperature of the wire W and the prior temperature difference are stored in the storage unit **67**. The wire temperature and the prior temperature difference that are stored in the storage unit **67** are changed by the control unit **68**.

(103) Also, in the power supply control processing, the control unit **68** increases the duty ratio of the PWM signal in a stepwise manner. A temperature difference table Q**1** is stored in the storage unit **67**. In the temperature difference table Q**1**, a plurality of upper limit values for the temperature difference between the wire temperature and the environmental temperature are shown in association with a plurality of duty ratios in PWM control. Every time the control unit **68** increases the duty ratio in PWM signal, the control unit **68** determines whether or not an anomaly has occurred in power supply to the load **13** through the wire W based on whether or not the temperature difference is the upper limit value, or more, of the temperature difference associated with the duty ratio of the PWM signal output to the driving circuit **31**.

(104) FIG. **5** is a diagram illustrating content of the temperature difference table Q**1**. As shown in FIG. **5**, temperature differences between the wire temperature and the environmental temperature are shown in the temperature difference table Q**1** in association with a plurality of duty ratios. In the example in FIG. **5**, a plurality of duty ratios are shown in increments of 10%. The upper limit values of the temperature difference associated with the respective duty ratios are shown. The upper limit value is a temperature difference calculated when the duty ratio of the PWM signal is adjusted to a duty ratio shown in the temperature difference table Q**1**, in the case where the power supply voltage of the DC power supply **12** is at a maximum value in a normal state, for example. In the temperature difference table Q**1**, as the duty ratio increases, the upper limit value increases.

(105) As described above, in the power supply control processing, the control unit **68** increases the duty ratio of the PWM signal in a stepwise manner. An increasing amount table Q**2** is stored in the storage unit **67**. In the increasing amount table Q**2**, a plurality of increasing amounts of the temperature difference between the wire temperature and the environmental temperature are shown in association with a plurality of amounts of increase in the duty ratio. Every time the control unit **68** increases the duty ratio of the PWM signal, the control unit **68** determines whether or not an anomaly has occurred in the power supply to the load **13** through the wire W based on whether or not the amount of increase in the temperature difference is the upper limit value of the increasing amount associated with the actual increase in the duty ratio or more.

(106) FIG. **6** is a diagram illustrating content of the increasing amount table Q**2**. As shown in FIG. **6**, in the increasing amount table Q**2**, upper limit values of the amount of increase in the temperature difference are shown in association with a plurality of amounts of increase in the duty ratio. In the example in FIG. **6**, the duty ratio is increased in increments of 10%. The upper limit

value is an amount of increase in the temperature difference calculated when the duty ratio of the PWM signal is increased as shown in the increasing amount table Q**2** in the case where the power supply voltage of the DC power supply **12** is at the largest value in a normal state, for example.
(107) Note that, when the environmental temperature is the same, the amount of increase in the temperature difference corresponds to the amount of increase in the wire temperature. The wire temperature is represented by the sum of the temperature difference and the environmental temperature. When the environmental temperature is the same, the amount of increase between two wire temperatures is represented by the amount of increase in the temperature difference.
(108) Vehicle Data Transmission Processing
(109) In the vehicle data transmission processing, the control unit **68** waits until vehicle data is input to the input unit **65** from the sensor **15***a*. When vehicle data is input to the input unit **65**, the control unit **68** acquires the vehicle data input to the input unit **65**. Next, the control unit **68** instructs the communicating unit **66** to transmit the acquired vehicle data to the integrated ECU **10**, and ends the vehicle data transmission processing. After ending the vehicle data transmission processing, the control unit **68** again executes the vehicle data transmission processing.
(110) Temperature Calculation Processing
(111) FIG. **7** is a flowchart illustrating a procedure of temperature calculation processing. As described above, the control unit **68** periodically executes the temperature calculation processing. In the temperature calculation processing, the control unit **68** acquires current value information indicating the wire current value of the wire W from the A/D conversion unit **62** (step S**1**). Once the output unit **64** outputs a duty ratio of the PWM signal, the control unit **68** acquires the current value information in a period in which the PWM signal is at a high-level voltage. Next, the control unit **68** reads out a prior temperature difference from the storage unit **67** (step S**2**). This prior temperature difference is a temperature difference calculated in the previous temperature calculation processing. In the temperature calculation processing that is first executed after the microcomputer **21** is started, the prior temperature difference is zero degrees. After executing step S**2**, the control unit **68** acquires environmental temperature information from the A/D conversion unit **63** (step S**3**).
(112) The control unit **68** calculates the temperature difference between the wire temperature and the environmental temperature by substituting a plurality of numerical values in the formulas [1] and [2] (step S**4**). The plurality of numerical values are a wire current indicated by the current value information acquired in step S**1**, the prior temperature difference read out in step S**2**, an environmental temperature indicated by the environmental temperature information acquired in step S**3**, and a duty ratio of the PWM signal output by the output unit **64**. When the output unit **64** outputs an off signal, the duty ratio is zero.
(113) Next, the control unit **68** changes the prior temperature difference stored in the storage unit **67** to the temperature difference calculated in step S**4** (step S**5**). The changed prior temperature difference is used next time temperature calculation processing is performed. The prior temperature difference is the latest temperature difference calculated in the temperature calculation processing. After executing step S**5**, the control unit **68** calculates the wire temperature by adding the temperature difference calculated in step S**4** to the environmental temperature indicated by the environmental temperature information acquired in step S**3** (step S**6**).
(114) Next, the control unit **68** changes the wire temperature stored in the storage unit **67** to the wire temperature calculated in step S**6** (step S**7**). Therefore, the wire temperature stored in the storage unit **67** is the latest wire temperature calculated in the temperature calculation processing. After executing step S**7**, the control unit **68** ends the temperature calculation processing.
(115) As described above, the prior temperature difference, which is the latest temperature difference, and the latest wire temperature are stored in the storage unit **67**.
(116) Power Supply Control Processing
(117) FIGS. **8** and **9** are flowcharts illustrating procedures of the power supply control processing.

In the power supply control processing, the control unit **68** adjusts the duty ratio of the PWM signal to a duty ratio with which the related value related to the load **13** matches a fixed target value. The related value is a wire current value, power supplied to the power, or the voltage value of a voltage applied to the load **13**. As described above, the wire current value is a current value of current flowing to the load **13** through the wire W.

(118) Also, a flag value indicating the state of the individual ECU **11***a* is stored in the storage unit **67**. The flag value is zero, one, or two, and is changed by the control unit **68**. The flag value of zero indicates that power is properly supplied to the load **13**. The flag value of one indicates that the current through the wire W has been forcibly cut off. The flag value of two indicates that an anomaly has occurred in power supply to the load **13**.

(119) The control unit **68** executes the power supply control processing in a state in which the output unit **64** outputs an off signal. In the power supply control processing, the control unit **68** first determines whether or not the load **13** is caused to start operating (step S**11**). In step S**11**, if the communication unit **66** has received instruction data for instructing that the load **13** be caused to start operating, the control unit **68** determines that the load **13** is to be caused to start operating. If the communication unit **66** has not received instruction data for instructing that the load **13** be caused to start operating, the control unit **68** determines that the load **13** is not to be caused to start operating. Upon determining that the load **13** is not to be caused to start operating (S**11**: NO), the control unit **68** again executes step S**11**, and waits until the communication unit **66** receives instruction data for instructing that the load **13** be caused to start operating.

(120) Upon determining that the load **13** is to be caused to start operating (S**11**: YES), the control unit **68** determines whether or not the flag value is zero (step S**12**). If it is determined that the flag value is zero (S**12**: YES), the control unit **68** acquires power supply voltage information from the A/D conversion unit **60** (step S**13**). Next, the control unit **68** calculates the duty ratio of the PWM signal with which the average value of the related value in a given period matches a fixed target value based on the power supply voltage indicated by the power supply voltage information acquired in step S**13** (step S**14**). The target value is set in advance. As described above, the given period is one cycle of the PWM signal, for example. For example, when the load **13** is a head light including LEDs (Light Emitting Diodes), the luminance of the load **13** increases as the average wire current in a given period increases. When the load **13** is a head light including LEDs, the related value is a wire current value. The target value is represented by a current value. The wire current flowing when the switch **30** is on is denoted as a switch current. The switch current is calculated based on the power supply voltage indicated by the power supply voltage information acquired in step S**13**. The duty ratio calculated by the control unit **68** in step S**14** is represented by a formula 100.Math.(target value)/(switch current value).

(121) For example, when the load **13** is a head light including an incandescent lamp, the luminance of the load **13** increases as the average value of power supplied to the load **13** in a given period increases. When the load **13** is a head light including an incandescent lamp, the related value is the power supplied to the load **13**. The target value is also represented by power. The power supplied to the load **13** when the switch **30** is on is denoted as load power. The load power is calculated based on the power supply voltage indicated by the power supply voltage information acquired in step S**13**. The duty ratio calculated by the control unit **68** in step S**14** is represented by a formula 100.Math.(target value)/(load power).

(122) For example, when the load **13** is a DC motor, the rotational speed of the load **13** increases as the average value of a voltage applied to the load **13** in a given period increases. When the load **13** is a DC motor, the related value is a voltage value of a voltage applied to the load **13**. The target value is also represented by a voltage value. The voltage applied to the load **13** when the switch **30** is on is denoted as a load voltage. The load voltage is calculated based on the power supply voltage indicated by the power supply voltage information acquired in step S**13**. The duty ratio calculated by the control unit **68** in step S**14** is represented by a formula 100.Math.(target value)/(load voltage

value).
(123) Next, the control unit **68** instructs the output unit **64** to output a PWM signal with the duty ratio calculated in step S**14** (step S**15**). With this, the driving circuit **31** performs PWM control on the switch **30** according to the PWM signal voltage. The duty ratio in the PWM control performed by the driving circuit **31** is adjusted to the duty ratio calculated in step S**14**. As a result of the driving circuit **31** performing PWM control on the switch **30**, a current flows through the wire W, and the average value of the related value is adjusted to the target value. As a result of current flowing through the wire W, power is supplied to the load through the wire W.
(124) Next, the control unit **68** reads out the latest wire temperature calculated in the temperature calculation processing from the storage unit **67** (step S**16**), and determines whether or not the read-out wire temperature is a wire temperature threshold or more (step S**17**). Upon determining that the wire temperature is less than the wire temperature threshold (S**17**: NO), the control unit **68** determines whether or not self-shutting off of the switch **30** has been performed by the driving circuit **31** (step S**18**). In step S**18**, if the largest voltage that is allowed to be output to the microcomputer **21** is input to at least one of the A/D conversion units **61** and **62**, the control unit **68** determines that the self-shutting off has been performed. If the maximum voltage is not input to the A/D conversion units **61** and **62**, the control unit **68** determines that the self-shutting off has not been performed.
(125) If it is determined that the self-shutting off has not been performed (S**18**: NO), the control unit **68** determines whether or not the load **13** is to be caused to stop operating (step S**19**). In step S**19**, if the communication unit **66** has received instruction data for instructing that the load **13** be caused to stop operating, the control unit **68** determines that the load **13** is to be caused to stop operating. If the communication unit **66** has not received instruction data for instructing that the load **13** be caused to stop operating, the control unit **68** determines that the load **13** is not to be caused to stop operating.
(126) Upon determining that the load **13** is not to be caused to stop operating (S**19**: NO), the control unit **68** acquires power supply voltage information from the A/D conversion unit **60** (step S**20**). Next, similarly to step S**14**, the control unit **68** calculates the duty ratio of the PWM signal with which the average value of the related value matches a target value based on the power supply voltage indicated by the power supply voltage information acquired in step S**20** (step S**21**). Next, the control unit **68** changes the duty ratio of the PWM signal output by the output unit **64** to the duty ratio calculated in step S**21** (step S**22**). After executing step S**22**, the control unit **68** again executes step S**16**.
(127) Assume that the wire temperature is less than the wire temperature threshold, self-shutting off has not been performed, and the communication unit **66** has not received instruction data instructing that the load **13** be caused to stop operating. In this case, the duty ratio is changed according to the power supply voltage of the DC power supply **12**. Specifically, when the power supply voltage decreases, the control unit **68** increases the duty ratio. When the power supply voltage increases, the control unit **68** reduces the duty ratio. Accordingly, even if the power supply voltage changes, the related value of the load **13** is kept at the target value. In the case where the positive electrode of the DC power supply **12** is connected to a starter of a vehicle C, when the starter has started operating, the power supply voltage decreases. When the starter has stopped operating in a similar case, the power supply voltage increases.
(128) Upon determining that the wire temperature is the wire temperature threshold or more (S**17**: YES), or determining that self-shutting off has been performed (S**18**: YES), the control unit **68** changes the flag value to one (step S**23**). Upon determining that the load **13** is to be caused to stop operating (S**19**: YES), or after executing step S**23**, the control unit **68** instructs the output unit **64** to output an off signal to the driving circuit **31** (step S**24**). With this, the driving circuit **31** keeps the switch **30** off. Accordingly, the current flowing through the wire W is cut off. As a result, power supply to the load **13** through the wire W stops. As described above, when the output unit **64**

outputs an off signal in a state in which the driving circuit **31** has performed self-shutting off, the self-shutting off is canceled.

(129) After executing step S**24**, the control unit **68** ends the power supply control processing. After ending the power supply control processing, the control unit **68** again executes the power supply control processing, and waits until the communication unit **66** receives instruction data for instructing that the load **13** be caused to start operating.

(130) Upon determining that the flag value is not zero (S**12**: NO), the control unit **68** determines whether or not the flag value is one (step S**25**). Upon determining that the flag value is one (step S**25**: YES), the control unit **68** determines whether or not the latest temperature difference calculated in the temperature calculation processing is a reference temperature difference or less (step S**26**). The reference temperature difference is a fixed value, and is zero degrees or a positive temperature close to zero degrees. The reference temperature difference is set in advance.

(131) Upon determining that the flag value is not one (S**25**: NO), or that the latest temperature difference exceeds the reference temperature difference (S**26**: NO), the control unit **68** ends the power supply control processing without causing the output unit **64** to output a PWM signal. After ending the power supply control processing, the control unit **68** again executes the power supply control processing.

(132) As described above, when the flag value is two, power supply to the load **13** cannot be started. In the case of the flag value being one as well, when the latest temperature difference exceeds the reference temperature difference, the control unit **68** does not instruct the output unit **64** to output a PWM signal.

(133) Upon determining that the latest temperature difference is the reference temperature difference or less (S**26**: YES), the control unit **68** instructs the output unit **64** to output a PWM signal to the driving circuit **31** (step S**27**). Accordingly, the driving circuit **31** performs PWM control. In step S**27**, the duty ratio of the PWM signal is adjusted to the smallest duty ratio among the plurality of duty ratios shown in the temperature difference table Q**1**. In the example in FIG. **5**, the duty ratio of the PWM signal is adjusted to 10%. As a result of the control unit **68** executing step S**27**, a current flows to the load **13** through the wire W. The duty ratio of the PWM signal is small, and therefore the average wire current value is small.

(134) After executing step S**27**, the control unit **68** determines whether or not the latest temperature difference calculated in the temperature calculation processing exceeds an upper limit value of the temperature difference associated with the duty ratio of the PWM signal output by the output unit **64**, in the temperature difference table Q**1** (step S**28**).

(135) Note that the control unit **68** executes step S**28** after a given period has elapsed since step S**27** was executed. The given period is one cycle of the temperature calculation processing or more. Therefore, the temperature difference is calculated at least once since step S**27** was executed.

(136) Upon determining that the latest temperature difference is the upper limit value or less (S**28**: NO), the control unit **68** determines whether or not the amount of increase in the temperature difference due to the increase in the duty ratio of the PWM signal exceeds the upper limit value of the increasing amount associated with the increase in the duty ratio that has been actually performed, in the increasing amount table Q**2** (step S**29**). In the example in FIG. **6**, when the duty ratio has increased from 0% to 10% in response to the output unit **64** outputting the PWM signal, it is determined whether or not the actual increasing amount exceeds the upper limit value of the increasing amount associated with the increase from 0% to 10%.

(137) Upon determining that the amount of increase in the temperature difference is the upper limit value or less (S**29**: NO), the control unit **68** determines whether or not the duty ratio of the PWM signal output by the output unit **64** is a reference duty ratio (step S**30**). The reference duty ratio is a fixed value and is set in advance. The reference duty ratio is the largest duty ratio shown in the temperature difference table Q**1**. In the example in FIG. **5**, the reference duty ratio is 100%.

(138) Note that the reference duty ratio is not limited to 100%. The reference duty ratio may also

be a value less than 100%. For example, when the largest value of the duty ratio of the PWM signal is 80%, in the case where the duty ratio of the PWM signal has been adjusted such that the average related value of the load **13** matches the target value, the reference duty ratio may be set to 80%.
(139) Upon determining that the duty ratio of the PWM signal is not the reference duty ratio (S**30**: NO), the control unit **68** increases the duty ratio of the PWM signal output by the output unit **64** (step S**31**). The duty ratio of the PWM signal output by the output unit **64** is denoted as an actual duty ratio. In step S**31**, the control unit **68** increases the duty ratio of the PWM signal to a duty ratio, in the temperature difference table Q**1**, that is larger than the actual duty ratio and is closest to the actual duty ratio. In the example in FIG. **5**, when the actual duty ratio is 10%, in step S**31**, the control unit **68** increases the duty ratio of the PWM signal to 20%.
(140) After executing step S**31**, the control unit **68** again executes step S**28**. The control unit **68** increases the duty ratio of the PWM signal, that is, the duty ratio in PWM control, in a stepwise manner, until the duty ratio of the PWM signal reaches the reference duty ratio. Accordingly, the average wire current in a given period increases in a stepwise manner. Every time the control unit **68** increases the duty ratio in PWM control, the control unit **68** executes steps S**28** and S**29**. Upon determining that the duty ratio of the PWM signal is the reference duty ratio (S**30**: YES), the control unit **68** changes the flag value to zero (step S**32**). After executing step S**32**, the control unit **68** executes step S**20**.
(141) As described above, if the state in which the temperature difference is the upper limit value or less, and the amount of increase in the temperature difference is the upper limit value or less continues during a period until the duty ratio of the PWM signal reaches the reference duty ratio, the control unit **68** determines that the individual ECU **11***a* is normal, and changes the flag value to zero. Thereafter, the control unit **68** executes step S**20**, and adjusts the duty ratio of the PWM signal according to the power supply voltage of the DC power supply **12**.
(142) Upon determining that the temperature difference exceeds the upper limit value (S**28**: YES), or determining that the amount of increase in the temperature difference exceeds the upper limit value (S**29**: YES), the control unit **68** instructs the output unit **64** to output an off signal to the driving circuit **31** (step S**33**). With this, the driving circuit **31** keeps the switch **30** off. After executing step S**33**, the control unit **68** changes the flag value to two (step S**34**), and ends the power supply control processing.
(143) As described above, when the temperature difference exceeds the upper limit value, or the amount of increase in the temperature difference exceeds the upper limit value, the control unit **68** determines that an anomaly has occurred in power supply to the load **13**, and changes the flag value to two. As described above, when the flag value is two, power is not be supplied to the load **13**. Steps S**28** and S**29** each correspond to processing for determining whether or not an anomaly has occurred in power supply to the load **13**. Therefore, determination as to whether or not an anomaly has occurred in power supply is made based on a comparison between the temperature difference and the upper limit value associated with the duty ratio in PWM control actually performed by the driving circuit **31**. Moreover, determination as to whether or not an anomaly has occurred in power supply is made based on the amount of increase in the temperature difference caused by the increase in duty ratio in PWM control.
(144) From the above, an anomaly in the power supply is a phenomenon in which the temperature difference increases to an upper limit value or more, or a phenomenon in which the amount of increase in the temperature difference (wire temperature) exceeds an upper limit value. An anomaly in the power supply occurs when two ends of the load **13** are shorted, for example.
(145) State Transition in Power Supply
(146) FIG. **10** is a state transition diagram of power supply. When power is normally supplied to the load **13**, the flag value is zero, and the power supply state is a normal state in which power is supplied normally. When the calculated wire temperature has increased to a temperature that is the wire temperature threshold or more, the control unit **68** instructs the driving circuit **31** to forcibly

switch the switch **30** off. When the wire current has increased to a current that is the current threshold or more, or when the switch temperature has increased to a temperature that is the switch temperature threshold or more, the driving circuit **31** forcibly switches the switch **30** off.
(147) By forcibly switching off the switch **30**, the current flowing through the wire W is forcibly cut off. As a result, the flag value is changed to one, and the power supply state transitions to a cutoff state in which the current flowing through the wire W is forcibly cut off. Even in a case where the power supply state is the normal state, there is a possibility that the power supply state will transition to the cutoff state due to external noise, for example.
(148) When an instruction that the load **13** be caused to start operating is made in a case where the power supply state is the cutoff state, the control unit **68** increases the duty ratio of the PWM signal in a stepwise manner. FIG. **11** is a diagram for describing a step-wise increase in the duty ratio of a PWM signal. Waveforms of the PWM signal that are different in the duty ratio are shown in FIG. **11**. As shown in FIG. **11**, the control unit **68** increases the duty ratio of the PWM signal in a stepwise manner. In the example in FIG. **11**, the duty ratio of the PWM signal increases from 10% to 50%, 80%, and 100% in this order.
(149) By increasing the duty ratio of the PWM signal in a stepwise manner, the wire current increases in a stepwise manner. Every time the control unit **68** increases the duty ratio, the control unit **68** determines whether or not an anomaly has occurred in power supply to the load **13** by performing a comparison between the temperature difference and the upper limit value and a comparison between the amount of increase in the temperature difference and the upper limit value. If the control unit **68** does not detect an anomaly in the power supply in a period until the duty ratio of the PWM signal has reached the reference duty ratio, the control unit **68** changes the flag value to zero, and causes the power supply state to transition to the normal state, as shown in FIG. **10**.
(150) If the control unit **68** has detected an anomaly in the power supply in a period until the duty ratio of the PWM signal has reached the reference duty ratio, the control unit **68** changes the flag value to two, and causes the power supply state to transition to an anomalous state in which an anomaly occurs in power supply to the load **13**. After the power supply state transitions to the anomalous state, the power supply state will not transition to other states, and power will not be supplied to the load **13**.
(151) Effects of individual ECU **11***a*
(152) In the individual ECU **11***a*, the control unit **68** increases the average wire current in a given period in a stepwise manner. Every time the control unit **68** increases the average wire current in a stepwise manner, the control unit **68** determines whether or not an anomaly has occurred in power supply based on the temperature difference between the wire temperature and the environmental temperature. Therefore, the control unit **68** can detect an anomaly in the power supply while the wire current is small. The control unit **68** increases the average wire current in a stepwise manner by increasing the duty ratio of the PWM signal (PWM control) in a stepwise manner. Therefore, a stepwise increase in the average wire current can be easily realized.
Second Embodiment
(153) In the first embodiment, power supply to the load **13** through the wire W is realized by the driving circuit **31** performing PWM control on the switch **30**. However, the method of realizing power supply to the load **13** is not limited to the method of the driving circuit **31** performing PWM control.
(154) The differences of the second embodiment from the first embodiment will be described below. The configurations other than the configurations described below are in common with the first embodiment. Therefore, the constituent units that are in common with the first embodiment are given the same reference numerals as the first embodiment, and the description thereof will be omitted.
(155) Configuration of Microcomputer **21**
(156) In the first embodiment, the output unit **64** shown in FIG. **4** outputs a PWM signal and an off

signal to the driving circuit **31** in accordance with an instruction from the control unit **68**. In the second embodiment, an output unit **64** also outputs an on signal indicating switching on of a switch **30** to the driving circuit **31**, in accordance with an instruction from a control unit **68**. The on signal is at a high-level voltage.
(157) Configuration of Switching Device **20**
(158) When the output unit **64** of a microcomputer **21** outputs an on signal to a driving circuit **31**, the driving circuit **31** switches the switch **30** on. While the microcomputer **21** is outputting an on signal, the driving circuit **31** keeps the switch **30** on. As described in the first embodiment, when the switch **30** is switched on, current flows from a positive electrode of a DC power supply **12** to the switch **30**, a wire W, and a load **13** in this order. With this, power is supplied to the load **13**, and the load **13** starts operating.
(159) When a wire current value is a current threshold or more in a case where the output unit **64** outputs a PWM signal or an on signal to the driving circuit **31**, the driving circuit **31** performs self-shutting off of the switch **30** irrelevant to the signal input to the driving circuit **31**. When the switch temperature of the switch **30** is a switch temperature threshold or more in a similar case, the driving circuit **31** performs self-shutting off of the switch **30** irrelevant to the signal input to the driving circuit **31**. When the driving circuit **31** has performed self-shutting off in relation to the wire current value or the switch temperature, the driving circuit **31** notifies the microcomputer **21** of the fact that self-shutting off has been performed, similarly to the first embodiment.
(160) Temperature Calculation Processing
(161) When the output unit **64** outputs an on signal, the control unit **68** calculates the wire temperature using the formulas [1] and [2], where the duty ratio D is set to 100%.
(162) Power Supply Control Processing
(163) FIG. **12** is a flowchart illustrating a procedure of power supply control processing in the second embodiment. A portion of the power supply control processing in the second embodiment is in common with a portion of the power supply control processing in the first embodiment. Steps S**11**, S**12**, S**16** to S**19**, and S**23** to S**34**, in the power supply control processing in the second embodiment, that are in common with those in the power supply control processing in the first embodiment will not be described in detail.
(164) In the power supply control processing in the second embodiment, upon determining that the flag value is zero (S**12**: YES), the control unit **68** instructs the output unit **64** to output an on signal (step S**41**). With this, the driving circuit **31** switches the switch **30** on. While the output unit **64** is outputting an on signal, the driving circuit **31** keeps the switch **30** on. As described above, when the switch **30** is on, power is supplied to the load **13**, and the load **13** operates. After executing step S**41**, the control unit **68** executes step S**16**.
(165) As described above, when the flag value is zero, power is supplied to the load **13** by keeping the switch **30** on.
(166) Upon determining that the load **13** is not to be caused to stop operating (S**19**: NO), the control unit **68** again executes step S**16**. Assume that the wire temperature is less than the wire temperature threshold, the self-shutting off has not been performed, and the communication unit **66** has not received instruction data instructing that the load **13** be caused to stop operating. In this case, the switch **30** is kept on.
(167) After executing step S**32**, the control unit **68** executes step S**41**. Therefore, after the flag value is changed from one to zero, that is, the power supply state has transitioned from the cutoff state to the normal state, the switch **30** is kept on.
(168) Effects of individual ECU **11***a*
(169) The individual ECU **11***a* in the second embodiment similarly achieves the effects achieved by the individual ECU **11***a* in the first embodiment, other than the effects obtained by changing the duty ratio in the PWM control according to the power supply voltage.
(170) Modifications

(171) The timing at which anomaly determination is made as to whether or not an anomaly has occurred in the power supply to the load **13** while increasing the wire current value in a stepwise manner, in the first and second embodiments, is not limited to the timing at which, when the power supply state is the cutoff state, the load **13** is caused to start operating. The timing at which the anomaly determination is made may be a timing at which the load **13** is first caused to start operating after the ignition switch of a vehicle C has been switched from off to on, for example. Also, the switch temperature detecting circuit **33** is not limited to a circuit in which the thermistor **50** is used. The switch temperature detecting circuit **33** needs only be a circuit that can detect the switch temperature of the switch **30**.
(172) The current value information output by the current detecting circuit **32** is not limited to being represented by a signal voltage according to the wire current, and may also be digital information, for example. The switch temperature information output by the switch temperature detecting circuit **33** is not limited to being represented by a signal voltage that changes according to the switch temperature, and may also be digital information, for example. Also, the method of making a notification that the self-shutting off has been executed is not limited to the method of applying a voltage, and may also be a method of outputting information indicating that the self-shutting off has been executed to the microcomputer **21**.
(173) The method of adjusting the wire current is not limited to the method of adjusting the duty ratio in the PWM control. When a variable resistor is disposed on a current path, the wire current may also be adjusted by adjusting the resistance of the variable resistor. The device that calculates a wire temperature is not limited to the individual ECU **11***a*. For example, the integrated ECU **10** may also calculate the wire temperature. The power supply control device that controls power supply is not limited to the individual ECU **11***a* that communicates with the integrated ECU **10**.
(174) The number of sensors that are connected to each of the individual ECU **11***a* and the plurality of individual ECUs **11***b* is not limited to one, and may also be two or more. The number of actuators **14** connected to each individual ECU **11***b* is not limited to one, and may also be two or more.
(175) The switch **30** is not limited to an N-channel FET, and may also be a semiconductor switch other than the N-channel FET. Examples of the semiconductor switch other than the N-channel FET include a P-channel FET, an IGBT (Insulated Gate Bipolar Transistor), a bipolar transistor, and the like.
(176) The first and second embodiments disclosed herein are illustrative in all aspects and should not be considered restrictive. The scope of the present disclosure is indicated by the scope of claims, not the above-mentioned meaning, and is intended to include all modifications within the meaning and scope equivalent to the scope of claims.

Claims

1. A power supply control device that controls power supply through a wire, comprising: a processing unit configured to execute processing, a switch disposed on a current path of current flowing through the wire; a switching circuit configured to switch the switch on or off; wherein the processing unit causes the switching circuit to perform PWM control in which the switch is alternately switched on and off, wherein a duty ratio of the PWM control is increased so as to increase an average value of a wire current value of a current flowing through the wire in a stepwise manner, and every time the average value of a wire current value is increased, determines whether or not an anomaly has occurred in power supply through the wire based on a temperature difference between a wire temperature of the wire and an environmental temperature in an area surrounding the wire.
2. The power supply control device according to claim 1, wherein the processing unit acquires the wire current value, calculates the temperature difference based on the acquired wire current value,

and every time the average value of the wire current value is increased, determines whether or not the anomaly has occurred based on a calculated temperature difference.
3. The power supply control device according to claim 2, wherein the switching circuit receives a signal, the switching circuit switches the switch on or off according to the received signal, the switching circuit, if the wire current value is a current threshold or more, or if the temperature of the switch is a switch temperature threshold or more, switches the switch off irrelevant to the received signal, and the processing unit increases the average value of a wire current value in a stepwise manner, after the switching circuit has switched the switch off irrelevant to the received signal.
4. The power supply control device according to claim 2, wherein the processing unit cuts off a flow of current flowing through the wire, when the wire temperature is a wire temperature threshold or more, and increases the average value of a wire current value in a stepwise manner, after the wire temperature has increased to a temperature that is the wire temperature threshold or more.
5. The power supply control device according to claim 1, further comprising: a storage unit that stores a plurality of upper limit values for the temperature difference in association with a plurality of duty ratios regarding the PWM control, wherein the processing unit determines, every time the average value of a wire current value is increased, whether or not the anomaly has occurred, based on whether or not the temperature difference exceeds an upper limit value associated with a duty ratio in the PWM control performed by the switching circuit.
6. The power supply control device according to claim 5, wherein the processing unit determines, every time the average value of a wire current value is increased, whether or not the anomaly has occurred, based on an amount of increase in the temperature difference that has increased due to the increase in the average value of a wire current value.
7. The power supply control device according to claim 5, wherein the switching circuit receives a signal, the switching circuit switches the switch on or off according to the received signal, the switching circuit, if the wire current value is a current threshold or more, or if the temperature of the switch is a switch temperature threshold or more, switches the switch off irrelevant to the received signal, and the processing unit increases the average value of a wire current value in a stepwise manner, after the switching circuit has switched the switch off irrelevant to the received signal.
8. The power supply control device according to claim 5, wherein the processing unit cuts off a flow of current flowing through the wire, when the wire temperature is a wire temperature threshold or more, and increases the average value of a wire current value in a stepwise manner, after the wire temperature has increased to a temperature that is the wire temperature threshold or more.
9. The power supply control device according to claim 1, wherein the processing unit determines, every time the average value of a wire current value is increased, whether or not the anomaly has occurred, based on an amount of increase in the temperature difference that has increased due to the increase in the average value of a wire current value.
10. The power supply control device according to claim 9, wherein the switching circuit receives a signal, the switching circuit switches the switch on or off according to the received signal, the switching circuit, if the wire current value is a current threshold or more, or if the temperature of the switch is a switch temperature threshold or more, switches the switch off irrelevant to the received signal, and the processing unit increases the average value of a wire current value in a stepwise manner, after the switching circuit has switched the switch off irrelevant to the received signal.
11. The power supply control device according to claim 9, wherein the processing unit cuts off a flow of current flowing through the wire, when the wire temperature is a wire temperature threshold or more, and increases the average value of a wire current value in a stepwise manner,

after the wire temperature has increased to a temperature that is the wire temperature threshold or more.

12. The power supply control device according to claim 1, wherein the switching circuit receives a signal, the switching circuit switches the switch on or off according to the received signal, the switching circuit, if the wire current value is a current threshold or more, or if the temperature of the switch is a switch temperature threshold or more, switches the switch off irrelevant to the received signal, and the processing unit increases the average value of a wire current value in a stepwise manner, after the switching circuit has switched the switch off irrelevant to the received signal.

13. The power supply control device according to claim 12, wherein the processing unit cuts off a flow of current flowing through the wire, when the wire temperature is a wire temperature threshold or more, and increases the average value of a wire current value in a stepwise manner, after the wire temperature has increased to a temperature that is the wire temperature threshold or more.

14. The power supply control device according to claim 1, wherein the processing unit cuts off a flow of current flowing through the wire, when the wire temperature is a wire temperature threshold or more, and increases the average value of a wire current value in a stepwise manner, after the wire temperature has increased to a temperature that is the wire temperature threshold or more.

15. An in-vehicle control device that controls operations of a load, comprising: a receiving unit configured to receive instruction data instructing that the load be caused to start or stop operating; a switch disposed on a current path of current flowing through the wire; a switching circuit configured to switch the switch on or off; a processing unit configured to execute processing, wherein the processing unit controls power supply to the load through a wire in accordance with instruction data received by the receiving unit, causes the switching circuit to perform PWM control in which the switch is alternately switched on and off, wherein a duty ratio of the PWM control is increased so as to increase an average value of a wire current value of a current flowing through the wire in a stepwise manner, and determines, every time the average value of a wire current value is increased, whether or not an anomaly has occurred in power supply through the wire, based on a temperature difference between a wire temperature of the wire and an environmental temperature in an area surrounding the wire.

16. A power supply control method of controlling power supply through a wire that includes steps to be executed by a computer, the method comprising: providing a switch on the wire, the switch configured to be turned on and off; a step of increasing, in a stepwise manner, an average value of a wire current value of a current flowing through the wire by increasing a duty ratio of a PWM control of the switch; and a step of determining, every time the average value of a wire current value is increased, whether or not an anomaly has occurred in power supply through the wire, based on a temperature difference between a wire temperature of the wire and an environmental temperature in an area surrounding the wire.